BASE CHANGE FOR THE UPPER SHRIEK OF A LOCALLY QUASI-FINITE MORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Dual base change for the lower shriek of a locally quasi-finite morphism over a cartesian square: g^{-1} f_! equals f'_! (g')^{-1} on abelian sheaves (more-etale, Lemma lemma-lqf-base-change-f-shriek, canonical tag 0F5J). cited at proof line 2522. The single named input: the dual lower-shriek base change statement from which the claim follows by uniqueness of adjoints.

Source reference. Original lemma 0F5J: sources/stacks-chow/tags.txt, source lines 18939–18939. Cited at proof line 2522: the corresponding dual statement for the f_! functors from which the claim follows.

Lemma 0.2. *Consider a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

of schemes with f locally quasi-finite. For any abelian sheaf $\mathcal{F}$ on $Y'_{étale}$ we have $(g')_*(f')^!\mathcal{F} = f^!g_*\mathcal{F}$.

Proof. By uniqueness of adjoint functors, this follows from the corresponding (dual) statement for the functors $f_!$. See Lemma 0F5J. □

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